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The gustatory-salivary reflex, the secretion of saliva from the salivary glands in response to gustation (taste), is primarily mediated by the parasympathetic division of the nervous system. In general, parasympathetic pathways originate from the brain stem or the sacral spinal cord. Fibers originating from nuclei in these regions synapse onto ganglia that lie near the target organ. Nuclei are nerve cell bodies clustered together in the central nervous system (CNS), and ganglia are nerve cell bodies clustered within the peripheral nervous system (PNS). Both preganglionic neurons (those originating from the nuclei) and postganglionic neurons (those originating from the ganglia) release acetylcholine (ACh).

In the gustatory-salivary reflex, activation of taste receptors in the tongue sends impulses to the superior and inferior salivatory nuclei in the brainstem. Local processing within these nuclei activates the parasympathetic pathways that innervate the salivary glands. Ultimately, ACh released from postganglionic neurons acts on muscarinic acetylcholine receptors (mAChRs) on the secretory cells of the salivary glands to stimulate salivation (Figure 1).

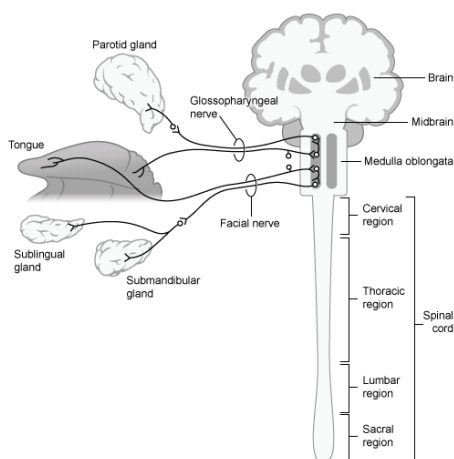


Figure 1 Parasympathetic regulation of salivary secretion

Sympathetic stimulation of the salivary glands also promotes salivation but to a lesser degree than parasympathetic stimulation. This may contribute to what is often perceived to be decreased salivation when the sympathetic division is more active than its counterpart. Unlike parasympathetic pathways (eg, gustatory-salivary reflex), sympathetic pathways originate in the thoracic and lumbar spinal segments, and the ganglia lie in close proximity to the spinal cord. Although preganglionic neurons of the sympathetic nervous system release ACh, the postganglionic neurons often release norepinephrine.

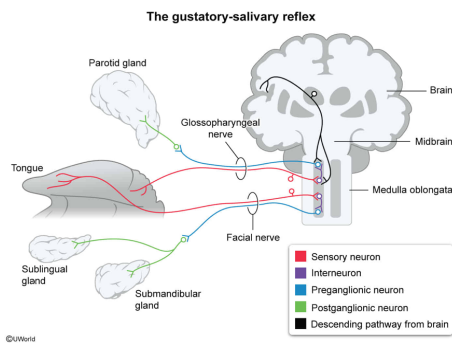
The gustatory-salivary reflex can be modulated (ie, dampened or enhanced) by input from higher centers in the CNS. To modulate the salivary response, descending pathways from the brain would most likely synapse onto which part of the reflex arc?

- ☐ A. The taste receptor [5%]
- ☐ B. The salivary glands [19%]
- ☒ C. The preganglionic neuron [38%]
- ☐ D. The postganglionic neuron [37%]

Correct

38% answered correctly

Explanation:



Reflexes are involuntary responses to stimuli that may or may not require input from the brain. A **reflex arc** is the specific neuronal pathway that facilitates the direct muscular or glandular effect associated with a particular reflex. Reflex arcs begin with the stimulation of a sensory neuron, which leads to an electrical impulse that travels toward the spine or brain along a sensory nerve. This impulse can be transmitted in **one of two ways**:

- **Directly** to the effector neuron (ie, a monosynaptic reflex arc), or
- **Indirectly** through an interneuron that interfaces with the effector neuron (ie, a polysynaptic reflex arc)

Subsequently, electrical impulses initiated in effector neurons leave the CNS and travel along an effector neuron axon to stimulate a **muscle fiber**, gland, or ganglionic neuron, either directly or after synapsing with a postganglionic effector neuron. When the effector neuron synapses with a muscle fiber, it is referred to as a *motor neuron*. Autonomic reflex arcs (eg, the gustatory-salivary reflex) are always polysynaptic.

Reflexes can be modulated (ie, dampened or enhanced) by input from the brain. In the case of polysynaptic reflexes, this modulation takes the form of descending signals from higher areas in the CNS that act on **preganglionic** neurons in the reflex arc. Based on the gustatory-salivary reflex arc depicted in Figure 1, the cell bodies of both the **interneuron** and the **preganglionic neuron** lie within the CNS and could be acted upon by descending input from higher brain centers to alter the gustatory-salivary reflex.

(Choices A and B) The taste receptor and salivary glands lie outside the spine (the tongue and oral cavity, respectively) and would not receive direct input from higher CNS areas.

(Choice D) By definition, a ganglion is a collection of neuron cell bodies that lie outside the CNS. Therefore, the cell body of a *postganglionic* neuron lies *outside* of the CNS and would *not* be directly acted upon by higher brain centers.

Educational objective:

A reflex is an involuntary response to a stimulus that may or may not require input from the brain. Reflexes are mediated by reflex arcs, neuronal pathways that include a sensory neuron, an effector neuron, and possibly an interneuron.

Time spent: 0 sec
Subject:
Biology

QID: 401535
System:
Endocrine and Nervous Systems

Saliva, a secreted fluid with multiple physiologic functions, is composed of water (~99%), proteins, and electrolytes. Together, these components facilitate nutrient consumption, respiration, and communication. For example, saliva aids in digestion by serving as a lubricant during mastication and by initiating carbohydrate and fat metabolism. In addition, calcium and phosphate present in saliva promote tooth remineralization, and antimicrobial and chemical buffering agents give saliva protective properties.

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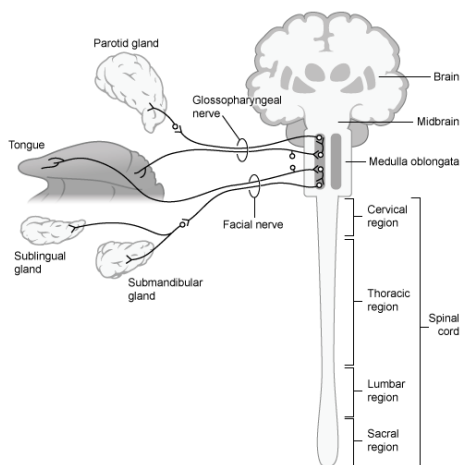


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Which of the following branches of the nervous system contain parts of the gustatory-salivary reflex pathway?

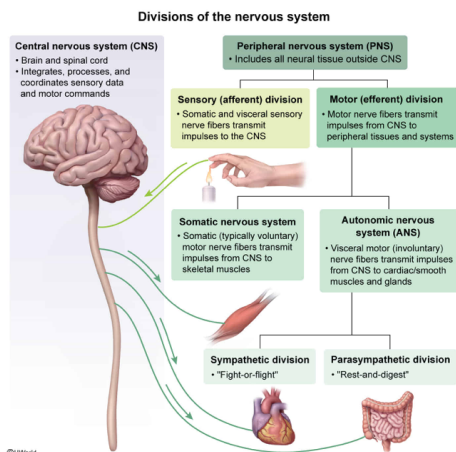
- I. Peripheral
- II. Somatic
- III. Autonomic

- ☐ A. I and II only [4%]
- ☒ B. I and III only [67%]
- ☐ C. III only [11%]
- ☐ D. I, II, and III [16%]

Correct

66% answered correctly

Explanation:



The nervous system is responsible for the control and integration of all body systems by receiving and processing incoming information from external and internal environments and coordinating purposeful reactions to this information. The nervous system can be broadly divided into two major components, the central and peripheral nervous systems. The **central nervous system (CNS)** is responsible for the integration of information and includes nuclei (bundles of cell bodies) and tracts (bundles of axons) within the *brain* and *spinal cord*.

In contrast, the **peripheral nervous system (PNS)** is composed of a *sensory division* and a *motor division*. The sensory (afferent) division functions to carry sensory impulses toward the CNS whereas the motor (efferent) division carries motor impulses away from the CNS. The PNS includes ganglia (bundles of cell bodies) and nerves (bundles of axons) that lie outside the CNS.

The **motor division of the PNS** can be **further divided** into the **somatic** and **autonomic** nervous systems. The somatic nervous system activates **skeletal muscle** responses, and the autonomic nervous system (ANS) activates involuntary smooth muscle, cardiac muscle, and glandular responses. Finally, the ANS is composed of the sympathetic and parasympathetic divisions. These two systems are considered antagonistic because they promote typically opposing effects.

The passage states that the **gustatory-salivary reflex** is part of the **parasympathetic** division of the nervous system. Therefore, the gustatory-salivary reflex is also part of the **peripheral** and autonomic branches of the nervous system (**Numbers I and III**).

(Number II) The somatic nervous system mediates skeletal muscle movement. Not all reflexes involve skeletal muscles, and the gustatory-salivary reflex pathway is one that does not modulate target tissue activity through skeletal muscle.

Educational objective:

The nervous system can be divided into two major branches, the central and peripheral nervous systems. The peripheral nervous system can be further divided into the sensory (afferent) and motor (efferent) divisions, with the motor division being composed of the somatic and autonomic nervous systems. Furthermore, the autonomic nervous system is divided into the parasympathetic and sympathetic branches.

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Subject:
Biology

QID: 401536
System:
Endocrine and Nervous Systems

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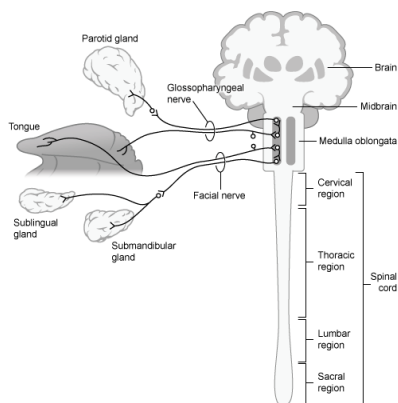


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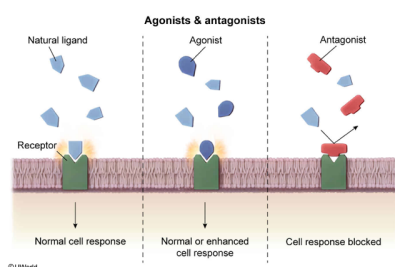
The drug cevimeline hydrochloride is a cholinergic agonist. Administration of cevimeline hydrochloride would be expected to cause:

- ☒ A. increased salivation due to stimulation of mAChRs. [79%]
- ☐ B. increased salivation due to inhibition of mAChRs. [1%]
- ☐ C. decreased salivation due to stimulation of mAChRs. [1%]
- ☐ D. decreased salivation due to inhibition of mAChRs. [17%]

Correct

78% answered correctly

Explanation:



Membrane-bound receptors respond to ligands that are either endogenous (originating from the organism) or exogenous (foreign to the organism). **Agonists** are ligands that activate the receptor and promote its downstream effects. In contrast, **antagonists** are ligands that deactivate or inhibit the receptor, preventing its downstream effects.

The defining characteristic of cholinergic signaling is that it involves acetylcholine (ACh). Cholinergic neurons are those that

secrete ACh, which binds to cholinergic receptors. According to the passage, parasympathetic pathways utilize ACh for signaling between the pre- and postganglionic neurons, as well as for signaling between the postganglionic neuron and the target tissue (eg, salivary glands). Because the gustatory-salivary reflex arc is mediated primarily by parasympathetic pathways, it must exert its effects (ie, stimulation of salivation) through *cholinergic* signaling.

The question states that **cevimeline hydrochloride** is a **cholinergic agonist** (ie, a ligand that *activates* ACh receptors). Treatment with a drug that **activates ACh receptors** (AChRs) would be expected to **increase salivation** by **activating muscarinic AChRs** (mAChRs) on secretory cells of the salivary glands (as well as AChRs on preganglionic sympathetic or parasympathetic neurons stimulating salivation).

(Choices B and D) Cevimeline hydrochloride is an *agonist* that *activates* mAChRs to stimulate salivation; it does *not* inhibit mAChRs.

(Choice C) Stimulation of mAChRs would lead to increased, *not decreased*, salivation. Inhibition of mAChRs by an antagonist would lead to decreased salivation.

Educational objective:

Agonists are ligands that activate the receptors they interact with and induce downstream effects. Antagonists are ligands that inhibit the receptors they interact with and block downstream effects.

Time spent:0 sec
Subject:
Biology

QID:401537
System:
Endocrine and Nervous Systems

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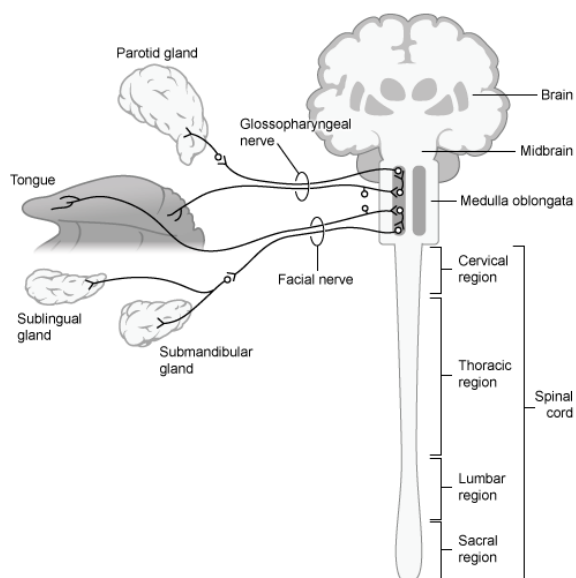


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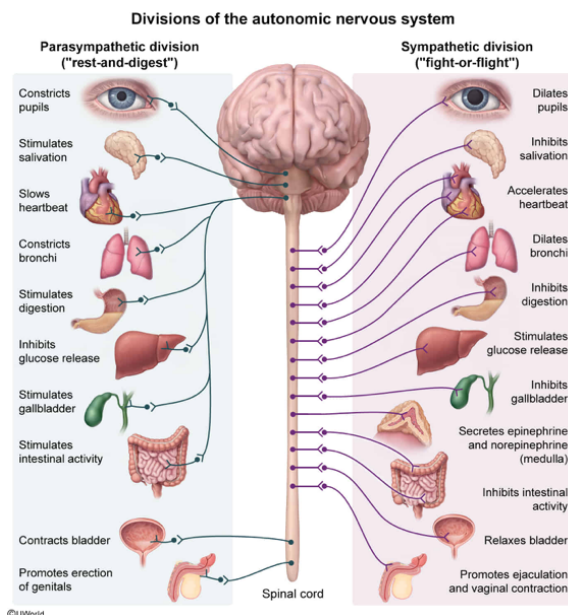
During periods of stress, many individuals experience dry mouth due to decreased salivation. Which of the following functions would also be expected to decrease during periods of stress?

- ☐ A. Gluconeogenesis [7%]
- ☐ B. Cardiac contraction [3%]
- ☐ C. Dilation of the pupils [8%]
- ✓ ☒ D. Digestive peristalsis [80%]

Correct

79% answered correctly

Explanation:



The autonomic nervous system includes two divisions, the parasympathetic and the sympathetic. In many cases, the parasympathetic and sympathetic divisions of the autonomic nervous system are considered antagonistic because they promote opposing effects.

The **parasympathetic ("rest and digest") division** promotes energy conservation and storage by lowering the heart rate, decreasing air flow through the lungs, stimulating digestion (eg, salivation, release of **digestive enzymes**, **peristalsis**), and promoting glucose uptake and **glycogenesis**.

In contrast, the **sympathetic ("fight or flight") division** mobilizes the body for immediate action against a threat. Oxygen delivery to the skeletal muscles is maximized (eg, increasing heart rate; dilating airways; constricting blood vessels that supply the visceral

organs, glands, and skin). Glycogen in the liver is broken down into glucose, which is then released into circulation to provide energy for skeletal muscles. The pupils dilate to maximize light intake by the eye, facilitating vision. Sympathetic signaling also inhibits nonessential functions such as those related to digestion and reproduction.

Acute and chronic stress results in activation of the sympathetic nervous system, and the passage states that salivation is perceived to decrease when sympathetic signaling is more active than parasympathetic signaling. In the given scenario, sympathetic activity is therefore expected to be higher than parasympathetic activity. As a result, individuals experiencing dry mouth due to stress (or anxiety) would also display a decrease in digestive activity, including but not limited to a **decrease in peristalsis** (ie, the involuntary wavelike contractions of smooth muscle that propels food through the digestive tract).

(Choices A, B, and C) Sympathetic activation would be expected to *increase* gluconeogenesis, cardiac contractility, and pupil dilation (diameter).

Educational objective:

The parasympathetic and sympathetic divisions of the nervous system are broadly antagonistic in their effects. The parasympathetic ("rest and digest") division promotes digestion and other tasks related to an organism's long-term needs. Alternatively, the sympathetic ("fight or flight") division enables the organism to meet more immediate needs.

Time spent:0 sec

Subject:
Biology

QID:401538

System:
Endocrine and Nervous Systems

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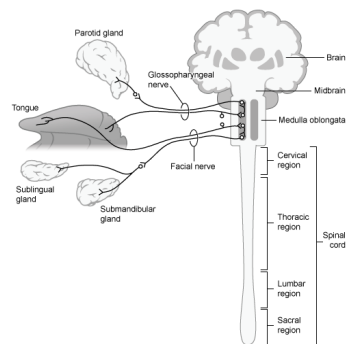


Figure 1 Parasympathetic regulation of salivary secretion

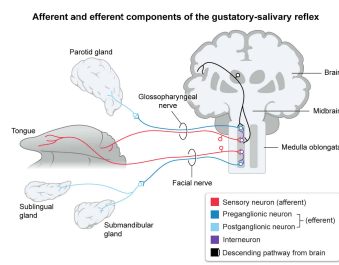
Sympathetic stimulation of the salivary glands also promotes salivation but to a lesser degree than parasympathetic stimulation. This may contribute to what is often perceived to be decreased salivation when the sympathetic division is more active than its counterpart. Unlike parasympathetic pathways (eg, gustatory-salivary reflex), sympathetic pathways originate in the thoracic and lumbar spinal segments, and the ganglia lie in close proximity to the spinal cord. Although preganglionic neurons of the sympathetic nervous system release ACh, the postganglionic neurons often release norepinephrine.

Which of the following neurons correspond to the gustatory-salivary reflex's afferent and efferent components, respectively?

- ☒ A. Sensory neuron, preganglionic neuron [69%]
- ☐ B. Preganglionic neuron, sensory neuron [12%]
- ☐ C. Interneuron, postganglionic neuron [14%]
- ☐ D. Postganglionic neuron, interneuron [3%]

Correct
68% answered correctly

Explanation:



The neural process of receiving and acting on sensory information in a reflex arc proceeds through the following steps:

1. An environmental event or disturbance (stimulus) is detected by **sensory receptors**, which are specialized nerve endings located throughout the body that recognize and respond to highly specific types of stimuli (eg, chemical, light, pressure, vibration).
2. The activated sensory neuron sends an *afferent* sensory signal (in the form of an electrical impulse) *toward* the CNS, where the signal is transmitted to an interneuron or effector neuron. (Note: *Afferent* information [approaches the CNS](#).)
3. As components of the CNS, **interneurons** in the brain and spinal cord act as **integration centers** that process and consolidate diverse sensory and CNS input from multiple locations. As such, interneurons can relay sensory information to the brain as well as integrate signals from the brain into the reflex response. However, because interneurons also transmit the signal directly to the effector neuron (and because interneurons are not present in every [reflex arc](#)), input from the brain is not always necessary for the completion of a reflex.
4. The effector neuron generates and sends an *efferent* signal that travels **toward** the target organ (typically, a muscle or gland). (Note: *Efferent* information *exits* the CNS.)
5. The **effector** (target) organ generates the desired response to the stimulus.

Within the gustatory-salivary reflex, the **sensory neuron** (which transmits impulses *toward* the CNS) makes up the *afferent* component of the reflex arc. The **pre- and postganglionic** fibers (which transmit impulses *away from* the spinal cord) make up the *efferent* component of the reflex arc.

(Choice B) The preganglionic neuron (which sends impulses *away* from the CNS) corresponds to the *efferent* component of the gustatory-salivary reflex. The sensory neuron (which sends impulses *toward* the CNS) corresponds to the *afferent* component of the gustatory-salivary reflex.

(Choices C and D) The interneuron propagates signals within the CNS but does not generate impulses that travel *directly* toward or away from the CNS. Therefore, the interneuron is neither afferent nor efferent.

Educational objective:

Afferent signals approach the CNS; efferent signals exit the CNS.

Time spent: 0 sec

Subject:
Biology

QID: 401539

System:
Endocrine and Nervous Systems

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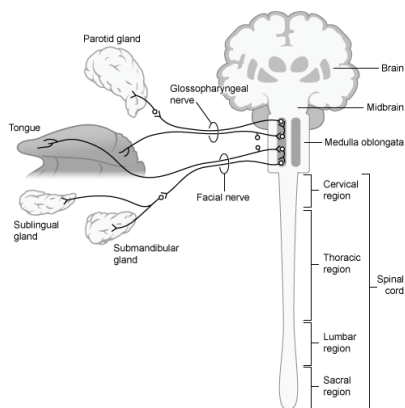
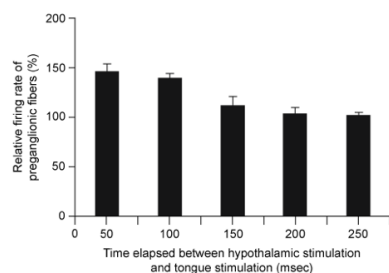


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The graph below shows the firing rate of preganglionic neurons within the rat gustatory-salivary reflex arc when the lateral hypothalamic area (LHA) is electrically stimulated at various intervals before the tongue is stimulated. The results are compared to the average preganglionic firing rate following tongue-only stimulation, which was used as a baseline that represented 100% activity. (Error bars indicate 95% confidence intervals; baseline error bars were similar to those at 250 msec.)



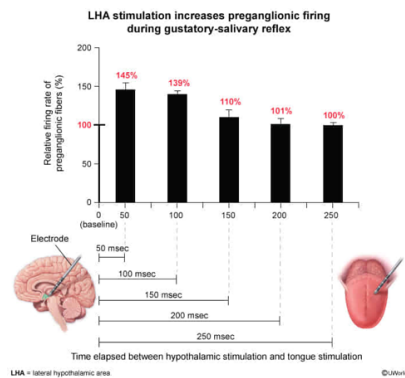
The results of this experiment indicate that at shorter time intervals:

- ☐ A. LHA signaling has no effect on the magnitude of preganglionic firing during the reflex. [6%]
- ☒ B. LHA signaling increases the magnitude of preganglionic firing during the reflex. [75%]
- ☐ C. LHA signaling decreases the magnitude of preganglionic firing during the reflex. [13%]
- ☐ D. There is not enough data to determine the effect of LHA signaling on the reflex. [5%]

Correct

75% answered correctly

Explanation:



In the scenario described in this question, researchers are trying to determine whether the lateral hypothalamic area (LHA) of the brain modulates the gustatory-salivary reflex response and, if it does, whether that modulation results in a diminished or enhanced reflex.

First, the researchers measured baseline activity in the preganglionic fiber of the gustatory-salivary parasympathetic reflex pathway, which was represented by the firing rate of the preganglionic fiber in response to tongue stimulation alone. The researchers then measured the activity of the fiber in response to sequential stimulation of the LHA and the tongue such that the LHA was stimulated at various intervals *before* the tongue.

According to the data, the preganglionic fiber responded to this sequential stimulation by exhibiting **significantly greater** activity at shorter intervals. This effect was most pronounced when the LHA was stimulated shortly prior to the tongue (<100 msec). The effect weakened as more time elapsed between LHA stimulation and tongue stimulation, and the firing rate of the fiber returned to baseline at a time difference of approximately 200 msec. Therefore, properly timed **LHA signaling increases** the activity of the preganglionic fiber of the gustatory-salivary pathway.

(Choice A) This graph demonstrates a clear relationship between the timing of LHA-tongue stimulation and the amount of activity within preganglionic fibers, with shorter intervals associated with significantly greater activity.

(Choice C) This graph shows that LHA-tongue stimulation significantly increases (*not* decreases) the activity of preganglionic fibers.

(Choice D) This graph indicates that signaling from the LHA does affect the gustatory-salivary reflex at some time points, based on the average values and associated confidence intervals. Therefore, enough information *is* provided to determine the effect of LHA signaling on the reflex.

Educational objective:

Input from higher areas in the CNS can modulate the activity of reflexes by either strengthening or weakening the magnitude of the response.

Time spent: 0 sec
Subject:
Biology

QID: 401540
System:
Endocrine and Nervous Systems